

HTML NOTES

1. Introduction to HTML

What is HTML?

HTML stands for **HyperText Markup Language**.

HTML is the standard markup language used to create and structure the content of web pages.

HTML is not a programming language because it does not contain programming logic such as loops, conditions, functions, or calculations. It is a **markup language** because we use predefined tags to tell the browser what each piece of content represents.

For example:

```
<h1>Welcome to My Website</h1>
<p>This is my first webpage.</p>
```

Here:

- `<h1>` tells the browser that the content is a main heading.
- `<p>` tells the browser that the content is a paragraph.

What does HyperText mean?

HyperText refers to text that can contain links to other documents, web pages, files, or resources.

For example:

```
<a href="https://www.google.com">Visit Google</a>
```

The text "Visit Google" becomes clickable and takes the user to another resource.

What does Markup Language mean?

Markup means using special tags to describe the structure and meaning of content.

For example:

```
<h1>HTML</h1>
<p>HTML is used to create web pages.</p>
```

The browser understands that:

- HTML is a heading.
 - The second line is a paragraph.
-

2. Basic HTML Document Structure

A basic HTML document looks like this:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>

<body>

  <h1>Hello World</h1>

</body>

</html>
```

Let's understand every part.

3. DOCTYPE

```
<!DOCTYPE html>
```

Definition

`<!DOCTYPE html>` tells the browser that the document is an HTML5 document.

It helps the browser render the page using modern HTML standards.

Important point

DOCTYPE is not an HTML tag.

It is a **document type declaration**.

Example

```
<!DOCTYPE html>
```

It should normally be the first line of an HTML document.

4. HTML Tag

```
<html lang="en">
```

The `<html>` element is the root element of an HTML document.

All other HTML elements are generally placed inside it.

Example:

```
<html>
  <head>
  </head>

  <body>
  </body>
</html>
```

lang attribute

```
<html lang="en">
```

`lang` specifies the primary language of the document.

For example:

```
<html lang="en">
```

means the document is primarily written in English.

Another example:

```
<html lang="fr">
```

indicates French.

5. Head Section

```
<head>
</head>
```

The `<head>` contains information about the webpage that is generally not displayed directly as page content.

Common elements inside `<head>` include:

- `<title>`
- `<meta>`
- `<link>`
- `<style>`
- `<script>`

Example:

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>My Website</title>
</head>
```

6. Meta Charset

```
<meta charset="UTF-8">
```

This specifies the character encoding used by the webpage.

UTF-8 supports a very large range of characters, including:

- English characters
- Numbers
- Special characters
- Many international languages
- Symbols

Example:

```
<meta charset="UTF-8">
```

It is commonly included in every HTML document.

7. Viewport Meta Tag

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

This is important for responsive web design.

It tells the browser how to control the page's dimensions and scaling on different devices.

Meaning

`width=device-width`

The webpage width should match the device's screen width.

`initial-scale=1.0`

The initial zoom level should be 1.

Without an appropriate viewport setting, webpages may not behave correctly on mobile devices.

8. Title Tag

`<title>Document</title>`

The `<title>` specifies the title of the webpage.

The title generally appears:

- In the browser tab
- In bookmarks
- In browser history
- In search engine results in many cases

Example:

`<title>Student Registration</title>`

9. Body Tag

`<body>`
`</body>`

The `<body>` contains the visible content of the webpage.

Examples of content placed inside `<body>`:

- Headings

- Paragraphs
- Images
- Links
- Tables
- Forms
- Audio
- Video
- Lists
- Buttons

Example:

```
<body>
```

```
  <h1>Welcome</h1>
```

```
  <p>This is my website.</p>
```

```
</body>
```

10. HTML Comments

HTML comments are written as:

```
<!-- This is a comment -->
```

Comments are not displayed on the webpage.

They are useful for:

- Explaining code
- Organizing code
- Temporarily disabling HTML code
- Leaving notes for developers

Example:

```
<!-- Student Information -->
```

```
<h1>Student Details</h1>
```

A complete section can also be temporarily commented:

```
<!--
```

```
<h1>Hello</h1>
```

```
<p>This content will not appear.</p>
```

```
-->
```

11. HTML Headings

HTML provides six heading levels:

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

Definition

Heading elements are used to represent headings and subheadings in a webpage.

`<h1>` represents the highest-level heading.

`<h6>` represents the lowest-level heading.

Example

```
<h1>Web Development</h1>
```

```
<h2>HTML</h2>
```

```
<h3>Introduction</h3>
```

```
<h2>CSS</h2>
```

```
<h3>Introduction</h3>
```

Use cases

Headings are useful for:

- Website titles
- Section headings
- Article headings
- Blog headings
- Product sections
- Documentation

Important point

Heading tags should be selected according to the **structure and importance of the content**, not simply according to how large the text should look.

CSS should be used when you want to control the visual size.

12. Paragraph Tag

The `<p>` tag is used to represent a paragraph.

Example:

```
<p>  
  HTML is used to create the structure of webpages.  
</p>
```

A paragraph can contain normal text and inline elements.

Example:

```
<p>  
  HTML is the structure of a webpage.  
</p>
```

Use cases

Paragraphs are commonly used for:

- Descriptions
 - Articles
 - Documentation
 - Introductions
 - Product information
 - Blog content
-

13. Formatting Tags

HTML provides several elements for giving meaning or emphasis to text.

Common formatting elements include:

```
<b>  
<strong>  
<i>  
<em>  
<u>
```

<ins>
<mark>

<sub>
<sup>

14. Bold Tag

HTML

 makes text visually bold.

Example:

<p>This is important text.</p>

Use case

Use when you want to draw visual attention to text without necessarily giving it additional importance or emphasis.

15. Strong Tag

Important

 indicates that the content has strong importance.

Example:

<p>
 Warning: Save your work before closing the browser.
</p>

Difference between and

 primarily represents stylistic attention.

 represents strong importance.

Example:

HTML

versus:

`<strong>Important Notice</strong>`

For meaningful emphasis, `<strong>` is generally more appropriate.

16. Italic Tag

`<i>HTML</i>`

`<i>` represents text in an alternate voice or mood, and browsers commonly display it in italic styling.

Example:

`<p>This is italic text.</p>`

It can be useful for:

- Technical terms
 - Foreign words
 - Titles
 - Alternate voice
-

17. Emphasis Tag

`<em>Important</em>`

`<em>` represents emphasized text.

Example:

`<p>`
You `<em>must</em>` submit the assignment today.
`</p>`

Difference between `<i>` and `<em>`

`<i>` represents alternate text or a different voice.

`<em>` represents emphasis.

18. Underline Tag

`<u>HTML</u>`

`<u>` represents text with an underline.

Example:

`<p>This is <u>underlined text</u>.</p>`

For styling purposes, CSS is generally preferred.

19. Inserted Text

`<ins>new content</ins>`

`<ins>` represents content that has been inserted into a document.

Example:

`<p>
The new price is <ins>₹999</ins>.
</p>`

Browsers commonly display inserted content with an underline.

20. Mark Tag

`<mark>Important</mark>`

`<mark>` represents highlighted or marked text.

Example:

`<p>
Remember the <mark>exam date</mark>.
</p>`

It is useful when you want to highlight a particular portion of content.

21. Delete Tag

`<del>1999</del>`

`<del>` represents content that has been deleted or removed from a document.

Example:

```
<p>  
  Old price: <del>₹1999</del>  
</p>
```

It is commonly displayed with a line through the text.

22. Subscript

`H<sub>2</sub>O`

`<sub>` represents subscript text.

Example:

```
<p>H<sub>2</sub>O</p>
```

Output represents:

H₂O

Use cases

Subscript is commonly used for:

- Chemical formulas
- Mathematical notation
- Scientific content

Example:

```
CO<sub>2</sub>
```

23. Superscript

`a<sup>2</sup>`

`<sup>` represents superscript text.

Example:

```
<p>a<sup>2</sup> + b<sup>2</sup></p>
```

It can be used for:

- Mathematical powers
- Footnote references
- Ordinal notation

Example:

```
x<sup>2</sup>
```

24. Line Break

The `<br>` element creates a line break.

Example:

```
<p>  
  HTML<br>  
  CSS<br>  
  JavaScript  
</p>
```

The content appears on separate lines.

Important point

`<br>` is a void element. It does not require a closing tag.

Correct:

```
<br>
```

Not normally written as:

```
<br></br>
```

Use cases

Useful when a line break is actually part of the content, such as:

- Addresses

- Poems
- Short lines of text

For layout purposes, CSS should generally be used instead.

25. Horizontal Rule

The `<hr>` element represents a thematic break between sections.

Example:

```
<p>Frontend Technologies</p>
```

```
<hr>
```

```
<p>Backend Technologies</p>
```

It is commonly displayed as a horizontal line.

Use cases

It can separate:

- Sections of an article
- Topics
- Content categories
- Different parts of a document

`<hr>` is also a void element.

26. HTML Lists

HTML provides three major types of lists:

1. Ordered List
 2. Unordered List
 3. Description List
-

27. Unordered List

An unordered list represents a list where the order of items is not important.

Syntax:

```
<ul>
  <li>Chilli</li>
  <li>Lemon</li>
  <li>Coriander</li>
</ul>
```

`<ul>` means unordered list.

`<li>` means list item.

The browser normally displays bullet points.

Use cases

Unordered lists are useful for:

- Features
- Ingredients
- Navigation menus
- Skills
- General collections of items

Example:

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

28. `type` Attribute in Unordered Lists

Example:

```
<ul type="circle">
  <li>Chilli</li>
  <li>Lemon</li>
  <li>Coriander</li>
</ul>
```

Historically, `type` could specify bullet styles such as:

```
<ul type="disc">
<ul type="circle">
<ul type="square">
```

However, modern HTML development generally uses CSS for list styling.

Example:

```
ul {
  list-style-type: circle;
}
```

29. Ordered List

An ordered list represents a sequence where order matters.

Syntax:

```
<ol>
  <li>Turn on the gas stove</li>
  <li>Put oil</li>
  <li>Add chilli</li>
</ol>
```

The browser normally displays:

1. Turn on the gas stove
2. Put oil
3. Add chilli

Use cases

Ordered lists are useful for:

- Steps
 - Instructions
 - Rankings
 - Procedures
 - Recipes
 - Top-N lists
-

30. Ordered List Type Attribute

The `type` attribute can specify the numbering format.

Example:

```
<ol type="i">
  <li>Step One</li>
  <li>Step Two</li>
</ol>
```

Possible traditional values include:

```
type="1"
type="A"
type="a"
type="I"
type="i"
```

Examples:

1, 2, 3

A, B, C

a, b, c

I, II, III

i, ii, iii

31. Start Attribute

The `start` attribute specifies the starting number.

Example:

```
<ol start="8">
  <li>Item</li>
  <li>Item</li>
  <li>Item</li>
</ol>
```

The list starts from 8.

32. Reversed Attribute

The `reversed` attribute makes an ordered list count backwards.

Example:

```
<ol reversed>
  <li>Step One</li>
  <li>Step Two</li>
  <li>Step Three</li>
</ol>
```

The browser displays the numbering in descending order.

33. Description List

A description list is used for terms and their descriptions.

It uses:

```
<dl>
  <dt>Term</dt>
  <dd>Description</dd>
</dl>
```

Example:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Elements

`<dl>` = Description list

`<dt>` = Description term

`<dd>` = Description/details of the term

Use cases

Description lists are useful for:

- Glossaries
 - Definitions
 - FAQs
 - Metadata
 - Terms and explanations
-

34. Anchor Tag

The `<a>` element is used to create hyperlinks.

Example:

```
<a href="https://www.google.com">
  Visit Google
</a>
```

The text becomes clickable.

Important attribute

`href` specifies the destination of the link.

Example:

```
<a href="https://www.example.com">
  Visit Website
</a>
```

35. Types of Links

External Link

```
<a href="https://www.google.com">
  Google
</a>
```

Points to another website.

Internal Link

```
<a href="about.html">
  About Us
</a>
```

Points to another page in the same website.

Email Link

```
<a href="mailto:test@example.com">
  Send Email
</a>
```

Opens the user's default email application.

Telephone Link

```
<a href="tel:+911234567890">
  Call Us
</a>
```

Useful especially on mobile devices.

Page Section Link

```
<a href="#contact">
  Contact
</a>
```

```
<h2 id="contact">Contact Us</h2>
```

Clicking the link takes the user to the element having `id="contact"`.

36. Opening Links in a New Tab

Example:

```
<a href="https://www.google.com" target="_blank">
  Google
</a>
```

`target="_blank"` generally opens the link in a new browsing context.

For external links, it is also common to use:

```
<a href="https://www.example.com"
  target="_blank"
  rel="noopener noreferrer">
  Visit Website
</a>
```

37. Images in HTML

The `<img>` element is used to display images.

Example:

```

```

Important attributes

`src`

Specifies the image source.

`alt`

Provides alternative text describing the image.

Example:

```

```

38. Why is `alt` Important?

The `alt` attribute is useful when:

- The image cannot load
- A screen reader is being used
- Accessibility is important
- The image conveys meaningful information

Example:

```

```

If the image cannot be displayed, the alternative text can provide information about it.

If an image is purely decorative, an empty alt can be appropriate:

```

```

39. Image Height and Width

Example:

```

```

You can specify dimensions using HTML attributes.

For modern layouts, CSS is generally preferred:

```
img {  
  width: 200px;  
  height: auto;  
}
```

This helps maintain the image's aspect ratio.

40. Audio Tag

The `<audio>` element is used to embed audio content.

Example:

```
<audio src="/Bayaan.mp3" controls></audio>
```

controls

Displays browser-provided audio controls.

The user can generally:

- Play
 - Pause
 - Change volume
 - Seek through the audio
-

41. Important Audio Attributes

controls

```
<audio src="song.mp3" controls></audio>
```

Displays controls.

autoplay

```
<audio src="song.mp3" autoplay></audio>
```

Requests automatic playback.

Modern browsers may restrict autoplay, especially when audio has sound.

loop

```
<audio src="song.mp3" loop controls></audio>
```

Repeats the audio after it ends.

muted

```
<audio src="song.mp3" muted></audio>
```

Starts the audio muted.

Example combining attributes:

```
<audio  
  src="/song.mp3"  
  controls  
  loop  
  muted>  
</audio>
```

42. Video Tag

The `<video>` element is used to embed video content.

Example:

```
<video  
  src="/dummyvdo.mp4"  
  controls  
  height="400">  
</video>
```

Common attributes

`src`

Specifies the video source.

`controls`

Displays video controls.

`height`

Controls the displayed height.

`width`

Controls the displayed width.

`poster`

Specifies an image displayed before the video starts.

`loop`

Repeats the video.

`autoplay`

Requests automatic playback.

`muted`

Starts the video without sound.

43. Poster Attribute

Example:

```
<video  
  src="./dummyvdo.mp4"  
  controls  
  poster="./images.webp">  
</video>
```

The `poster` image is displayed before the video starts playing.

Use cases

A poster can be used as:

- Video thumbnail
 - Preview image
 - Course video cover
 - Advertisement preview
-

44. HTML Tables

Tables are used to represent data in rows and columns.

Basic example:

```
<table border="1">

  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>Role</th>
  </tr>

  <tr>
    <td>Saomya</td>
    <td>25</td>
    <td>Trainer</td>
  </tr>

</table>
```

45. Table Elements

HTML tables commonly use:

```
<table>
<tr>
<th>
<td>
<thead>
<tbody>
<tfoot>
```

46. Table Tag

```
<table>  
</table>
```

Defines the table.

Example:

```
<table>  
  ...  
</table>
```

47. Table Row

```
<tr>  
</tr>
```

tr means Table Row.

Example:

```
<tr>  
  <td>Saomya</td>  
  <td>25</td>  
</tr>
```

This creates one row.

48. Table Header

```
<th>Name</th>
```

th means Table Header.

It represents a heading for a column or row.

Example:

```
<tr>  
  <th>Name</th>
```

```
<th>Course</th>
<th>Marks</th>
</tr>
```

49. Table Data

```
<td>Saomya</td>
```

td means Table Data.

It represents an individual data cell.

Example:

```
<tr>
  <td>Saomya</td>
  <td>CSE</td>
  <td>90</td>
</tr>
```

50. Thead

```
<thead>
</thead>
```

<thead> groups the header rows of a table.

Example:

```
<table>

  <thead>
    <tr>
      <th>Name</th>
      <th>Course</th>
    </tr>
  </thead>

</table>
```

51. Tbody

```
<tbody>
</tbody>
```

<tbody> groups the main data rows.

Example:

```
<tbody>
  <tr>
    <td>Saomya</td>
    <td>CSE</td>
  </tr>

  <tr>
    <td>Sakshi</td>
    <td>ECE</td>
  </tr>
</tbody>
```

52. Tfoot

```
<tfoot>
</tfoot>
```

<tfoot> groups footer rows.

It is useful for:

- Totals
- Averages
- Summaries
- Calculations

Example:

```
<tfoot>
  <tr>
    <td colspan="2">Average</td>
    <td>65.66</td>
  </tr>
</tfoot>
```

53. Complete Table Example

```
<table border="1">

  <thead>
    <tr>
      <th>Name</th>
      <th>Course</th>
      <th>Marks</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>Saomya</td>
      <td>CSE</td>
      <td>90</td>
    </tr>

    <tr>
      <td>Sakshi</td>
      <td>CSE</td>
      <td>79</td>
    </tr>

    <tr>
      <td>Tanya</td>
      <td>BBA</td>
      <td>98</td>
    </tr>
  </tbody>

  <tfoot>
    <tr>
      <td colspan="2">Average</td>
      <td>89</td>
    </tr>
  </tfoot>

</table>
```

54. Colspan

colspan is used to make one cell span across multiple columns.

Example:

```
<td colspan="2">
  Average
</td>
```

If there are two columns, this cell occupies the space of both columns.

Example:

```
<table border="1">

  <tr>
    <td colspan="3">Student Details</td>
  </tr>

  <tr>
    <td>Name</td>
    <td>Course</td>
    <td>Marks</td>
  </tr>

</table>
```

55. Rowspan

rowspan makes a cell span across multiple rows.

Example:

```
<table border="1">

  <tr>
    <td rowspan="3">Java</td>
    <td>Ram</td>
  </tr>

  <tr>
    <td>Priya</td>
  </tr>

  <tr>
    <td>Shyam</td>
  </tr>

</table>
```

Here, the Java cell occupies three rows.

56. Difference Between Colspan and Rowspan

colspan

Spans multiple columns horizontally.

```
<td colspan="3">
```

rowspan

Spans multiple rows vertically.

```
<td rowspan="3">
```

A simple way to remember:

colspan = multiple columns

rowspan = multiple rows

57. Cellspacing and Cellpadding

Older HTML table examples often use:

```
<table border="1" cellspacing="0" cellpadding="10">
```

Cellspacing

Controls the space between table cells.

Cellpadding

Controls the space between a cell's content and its border.

These attributes are considered legacy presentation techniques. Modern development generally uses CSS.

Example:

```
table {
```

```
border-collapse: collapse;
}

td, th {
padding: 10px;
}
```

58. Border Attribute

Example:

```
<table border="1">
```

Historically, the **border** attribute was used to display table borders.

Modern HTML generally uses CSS instead.

Example:

```
table,
th,
td {
border: 1px solid black;
}
```

59. Forms in HTML

A form is used to collect information from users.

Examples:

- Login form
- Registration form
- Contact form
- Feedback form
- Search form
- Payment form
- Job application form

Basic structure:

```
<form>
```

```
<label>Username</label>
```

```
<input type="text">
```

```
<input type="submit">
```

```
</form>
```

60. Form Element

```
<form>
```

```
</form>
```

The `<form>` element represents a section containing interactive controls for submitting information.

Example:

```
<form>
```

```
  <input type="text">
```

```
  <input type="submit">
```

```
</form>
```

61. Label

The `<label>` element provides a text label for a form control.

Example:

```
<label for="username">
```

```
  Username
```

```
</label>
```

```
<input type="text" id="username">
```

The `for` attribute of the label should match the `id` of the corresponding form control.

Here:

```
for="username"
```

matches:

```
id="username"
```

This establishes a relationship between the label and input.

62. Why Use Label?

Labels improve:

- Accessibility
- Usability
- Form clarity

Users can also click the label to focus or activate the associated control in many cases.

Example:

```
<label for="email">Email</label>  
<input type="email" id="email">
```

63. Text Input

```
<input type="text">
```

Used to collect single-line text.

Example:

```
<input  
  type="text"  
  placeholder="Enter your username">
```

Common use cases

- Username
 - Name
 - City
 - Address
 - Search keywords
-

64. Placeholder

Example:

```
<input
  type="text"
  placeholder="Enter username">
```

placeholder displays temporary hint text inside the input.

Important distinction:

Placeholder is not a replacement for a label.

Good form:

```
<label for="username">Username</label>
```

```
<input
  id="username"
  type="text"
  placeholder="Enter username">
```

65. Password Input

```
<input type="password">
```

Used to collect password-like information.

Example:

```
<label for="password">Password</label>
```

```
<input
  id="password"
  type="password">
```

The browser hides the entered characters visually.

66. Email Input

```
<input type="email">
```

Used to collect email addresses.

Example:

```
<label for="email">Email</label>
```

```
<input
  id="email"
  type="email"
  placeholder="Enter your email">
```

Browsers can perform built-in validation for email-like input.

67. Number Input

```
<input type="number">
```

Used for numeric input.

Example:

```
<input
  type="number"
  placeholder="Enter age">
```

Useful for:

- Age
- Quantity
- Number of items
- Numeric values

Additional attributes can restrict values:

```
<input
  type="number"
  min="1"
  max="100">
```

68. Date Input

```
<input type="date">
```

Allows the user to select a date.

Example:

```
<label for="dob">Date of Birth</label>
```

```
<input
  id="dob"
  type="date">
```

69. Datetime-local Input

```
<input type="datetime-local">
```

Allows the user to select a local date and time.

Example:

```
<input type="datetime-local">
```

Useful for:

- Appointment scheduling
 - Event scheduling
 - Meeting creation
 - Booking systems
-

70. URL Input

```
<input type="url">
```

Used for entering a URL.

Example:

```
<label for="website">Website</label>
```

```
<input
  id="website"
  type="url"
  placeholder="https://example.com">
```

Browsers can provide built-in validation for URL-like values.

71. Search Input

```
<input type="search">
```

Used specifically for search terms.

Example:

```
<input
  type="search"
  placeholder="Search products">
```

Commonly used in:

- Search bars
 - Product search
 - Website search
 - Filtering interfaces
-

72. File Input

```
<input type="file">
```

Allows the user to select a file from their device.

Example:

```
<input type="file">
```

73. Accept Attribute

Example:

```
<input
  type="file"
  accept=".mp3,.pdf">
```

The **accept** attribute provides a hint about which file types the user should select.

Example:

```
<input
  type="file"
  accept="image/*">
```

This indicates that image files are expected.

Another example:

```
<input
  type="file"
  accept=".pdf">
```

This indicates PDF files.

74. Multiple Attribute

Example:

```
<input
  type="file"
  multiple>
```

Allows the user to select multiple files.

Example:

```
<input
  type="file"
  accept=".pdf,.mp3"
  multiple>
```

This allows multiple selected files of the specified types.

75. Color Input

```
<input type="color">
```

Allows the user to select a color.

Example:

```
<label for="color">Choose a color</label>
```

```
<input
  id="color"
  type="color">
```

Use cases:

- Theme customization
- Drawing applications

- Design tools
 - Color pickers
-

76. Radio Button

```
<input type="radio">
```

A radio button is used when the user should select one option from a group of related choices.

Example:

```
<label>  
  <input type="radio" name="gender" value="male">  
  Male  
</label>
```

```
<label>  
  <input type="radio" name="gender" value="female">  
  Female  
</label>
```

Important concept: **name**

Radio buttons must generally have the same **name** to behave as one group.

Example:

```
<input type="radio" name="course" value="java">  
<input type="radio" name="course" value="python">  
<input type="radio" name="course" value="javascript">
```

The user can select one option from that group.

77. Checkbox

```
<input type="checkbox">
```

A checkbox represents an independent option that can be selected or unselected.

Example:

```
<label>
```

```
<input type="checkbox">  
I agree to the terms and conditions.  
</label>
```

Multiple checkboxes can be selected.

Example:

```
<label>  
  <input type="checkbox" name="skill" value="html">  
  HTML  
</label>
```

```
<label>  
  <input type="checkbox" name="skill" value="css">  
  CSS  
</label>
```

```
<label>  
  <input type="checkbox" name="skill" value="javascript">  
  JavaScript  
</label>
```

78. Radio Button vs Checkbox

Radio button

Generally used when selecting one option from a group.

Example:

Choose your gender:

() Male

() Female

Checkbox

Used when multiple independent choices can be selected.

Example:

Select your skills:

[] HTML

[] CSS

[] JavaScript

79. Week Input

```
<input type="week">
```

Allows the user to select a week.

Example:

```
<label for="week">Select Week</label>
```

```
<input  
  id="week"  
  type="week">
```

80. Month Input

```
<input type="month">
```

Allows the user to select a month and year.

Example:

```
<input type="month">
```

Useful for:

- Billing periods
 - Reports
 - Subscription periods
 - Monthly schedules
-

81. Range Input

```
<input type="range">
```

Creates a slider control.

Example:

```
<label for="volume">Volume</label>
```

```
<input  
  id="volume"  
  type="range">
```

You can control its range using:

```
<input
  type="range"
  min="0"
  max="100"
  value="50">
```

This creates a slider ranging from 0 to 100 with an initial value of 50.

82. Submit Input

```
<input type="submit">
```

Creates a button that submits the form.

Example:

```
<form>
```

```
  <input type="text">
```

```
  <input type="submit">
```

```
</form>
```

The default displayed text may be something like "Submit".

You can customize it using:

```
<input
  type="submit"
  value="Register">
```

83. Button Element

Another way to create a button is:

```
<button>Submit</button>
```

The `<button>` element is more flexible because it can contain text and, depending on the use case, other phrasing content.

Example:

```
<button type="submit">
  Register
</button>
```

Common button types include:

```
<button type="submit">Submit</button>
```

```
<button type="reset">Reset</button>
```

```
<button type="button">Click Me</button>
```

84. Select Element

The `<select>` element creates a dropdown list.

Example:

```
<select>
  <option>Car</option>
  <option>Bus</option>
  <option>Bike</option>
</select>
```

The user can choose one option.

85. Option Element

`<option>` represents an individual choice inside a `<select>`.

Example:

```
<select>
  <option>Car</option>
  <option>Bus</option>
  <option>Bike</option>
</select>
```

Each `<option>` represents one possible value.

86. Selected Attribute

You can make an option selected by default.

Example:

```
<select>
  <option>Car</option>
  <option selected>Bus</option>
  <option>Bike</option>
</select>
```

Bus will initially be selected.

87. Textarea

The `<textarea>` element is used to collect multi-line text.

Example:

```
<textarea></textarea>
```

A better example:

```
<label for="message">Message</label>
```

```
<textarea
  id="message"
  rows="5"
  cols="30"
  placeholder="Enter your message">
</textarea>
```

Use cases

Textarea is useful for:

- Feedback
 - Comments
 - Messages
 - Addresses
 - Descriptions
 - Long-form responses
-

88. Rows and Columns in Textarea

Example:

```
<textarea  
  rows="5"  
  cols="30">  
</textarea>
```

rows controls the visible number of text lines.

cols controls the approximate visible character width.

In modern responsive designs, CSS is often preferred for sizing.

89. Important Form Attributes

Several attributes are commonly used with form controls.

id

Uniquely identifies an element.

```
<input id="username">
```

Useful for:

- Connecting labels
 - JavaScript DOM selection
 - CSS selectors
-

name

Specifies the name associated with a form control.

```
<input  
  type="text"  
  name="username">
```

The **name** is particularly important when form data is submitted.

value

Represents the control's value.

Example:

```
<input  
  type="text"  
  value="Saomya">
```

For radio buttons and checkboxes, **value** identifies the submitted choice.

required

Makes a field mandatory.

Example:

```
<input  
  type="email"  
  required>
```

The browser will prevent normal form submission until the required field is appropriately filled.

disabled

Disables a form control.

Example:

```
<input  
  type="text"  
  disabled>
```

The user cannot normally interact with the disabled control.

readonly

Makes a text-like control read-only.

Example:

```
<input
  type="text"
  value="Saomya"
  readonly>
```

The user can generally see and select the value but cannot edit it.

checked

Used for checkbox and radio controls.

Example:

```
<input
  type="checkbox"
  checked>
```

The checkbox is selected initially.

selected

Used for `<option>`.

Example:

```
<option selected>JavaScript</option>
```

90. Complete Form Example

A practical student registration form can look like this:

```
<form>
```

```
  <label for="studentName">
    Student Name
  </label>
```

```
<input
  id="studentName"
  name="studentName"
  type="text"
  placeholder="Enter your name"
  required>

<br><br>

<label for="email">
  Email
</label>

<input
  id="email"
  name="email"
  type="email"
  placeholder="Enter your email"
  required>

<br><br>

<label for="age">
  Age
</label>

<input
  id="age"
  name="age"
  type="number"
  min="1"
  max="100">

<br><br>

<label for="course">
  Course
</label>

<select id="course" name="course">

  <option value="html">
    HTML
  </option>

  <option value="css">
    CSS
  </option>
```

```
    <option value="javascript">
      JavaScript
    </option>
```

```
</select>
```

```
<br><br>
```

```
<p>Gender</p>
```

```
<label>
  <input
    type="radio"
    name="gender"
    value="male">
```

```
    Male
```

```
</label>
```

```
<label>
  <input
    type="radio"
    name="gender"
    value="female">
```

```
    Female
```

```
</label>
```

```
<br><br>
```

```
<label>
  <input
    type="checkbox"
    name="terms"
    required>
```

```
    I agree to the terms.
```

```
</label>
```

```
<br><br>
```

```
<label for="message">
  Message
</label>
```

```
<br>
```

```
<textarea
  id="message"
  name="message"
  rows="5"
  cols="30"
  placeholder="Enter your message">
</textarea>

<br><br>

<button type="submit">
  Register
</button>

</form>
```

91. Important Concept: HTML Attributes

Attributes provide additional information or configuration for an HTML element.

General syntax:

```
<tag attribute="value">
```

Example:

```
<input
  type="text"
  id="username"
  placeholder="Enter username">
```

Here:

```
type
id
placeholder
```

are attributes.

Their corresponding values are:

```
text
username
Enter username
```

92. Common HTML Attributes

Some commonly used attributes are:

- id
- class
- title
- style
- href
- src
- alt
- target
- type
- name
- value
- placeholder
- required
- disabled
- readonly
- checked
- selected
- min
- max
- multiple
- accept

The exact attributes available depend on the HTML element.

93. Global Attributes

Some attributes can be used on many HTML elements.

Examples:

id

```
<p id="intro">  
  Welcome  
</p>
```

class

```
<p class="description">  
  Welcome
```

</p>

title

```
<p title="Additional information">  
  Hover over me  
</p>
```

hidden

```
<p hidden>  
  This is hidden.  
</p>
```

style

```
<p style="color: red;">  
  Hello  
</p>
```

For larger applications, CSS stylesheets are generally preferred over inline styles.

94. Block-Level and Inline Elements

HTML elements can behave differently in the layout.

Block-level behavior

A block-level element generally starts on a new line and takes up the available horizontal space.

Examples commonly include:

```
<div>  
<p>  
<h1>  
<h2>  
<section>  
<table>  
<form>  
<ul>  
<ol>
```

Example:

```
<p>First paragraph</p>
<p>Second paragraph</p>
```

The paragraphs appear on separate lines.

95. Inline Elements

Inline elements generally occupy only the space required by their content.

Examples commonly include:

```
<a>
<b>
<strong>
<i>
<em>
<mark>
<span>
```

Example:

```
<p>
  This is <strong>important</strong> information.
</p>
```

The `<strong>` element stays within the same line as the paragraph content.

96. Semantic Meaning

HTML is not only about making content appear on the screen.

Modern HTML encourages developers to use elements according to their meaning.

For example:

```
<header>
<nav>
<main>
<section>
<article>
<aside>
<footer>
```

These elements help communicate the structure and meaning of a webpage.

Example:

```
<header>
  <h1>My Website</h1>
</header>

<nav>
  <a href="/">Home</a>
  <a href="/about.html">About</a>
</nav>

<main>
  <section>
    <h2>Courses</h2>
    <p>Learn web development.</p>
  </section>
</main>

<footer>
  Copyright 2026
</footer>
```

Semantic HTML is beneficial for:

- Accessibility
- Search engines
- Maintainability
- Code readability
- Page structure

Quick HTML Tag Reference

Document Structure

```
<!DOCTYPE html>
<html>
<head>
<body>
<title>
<meta>
```

Text

<h1>
<h2>
<h3>
<h4>
<h5>
<h6>
<p>

<hr>

Formatting

<i>

<u>
<ins>
<mark>

<sub>
<sup>

Lists

<dl>
<dt>
<dd>

Links and Media

<a>

<audio>
<video>

Tables

<table>
<thead>
<tbody>
<tfoot>
<tr>
<th>

<td>

Forms

<form>

<label>

<input>

<select>

<option>

<textarea>

<button>

101. Important HTML Input Types

The most commonly used input types include:

<input type="text">

<input type="password">

<input type="email">

<input type="number">

<input type="date">

<input type="datetime-local">

<input type="url">

<input type="search">

<input type="file">

<input type="color">

<input type="radio">

<input type="checkbox">

<input type="week">

<input type="month">

<input type="range">

<input type="submit">

<input type="reset">

<input type="button">

102. Real-World Use Cases

HTML is used in almost every website and web application.

E-commerce website

HTML can create:

Product title
Product image
Product description
Price
Product table
Forms
Buttons
Links

Banking website

HTML can create:

Login form
Account details
Transaction tables
Navigation
Buttons
Forms

Educational website

HTML can create:

Course pages
Student registration
Video content
Audio lessons
Tables
Assignments
Forms

Social media website

HTML can create:

Posts
Images
Videos
Comments
Forms
Navigation
Profiles

103. Best Practices

Use meaningful HTML

Prefer:

```
<h1>Student Registration</h1>
```

instead of using a random element only because it looks large.

Use labels with form fields

Prefer:

```
<label for="email">Email</label>  
<input id="email" type="email">
```

Always provide useful alt text for meaningful images

```

```

Use semantic elements where appropriate

Prefer:

```
<header>  
<nav>  
<main>  
<section>  
<footer>
```

when they accurately represent the content.

Use CSS for presentation

Instead of depending on old presentation attributes such as:

```
<table border="1">
```

use CSS:

```
table {  
    border: 1px solid black;  
}
```

Keep HTML responsible for structure

Do not use HTML merely to achieve visual styling. Use CSS for visual design.

104. Most Important Interview Questions

What is HTML?

HTML is a markup language used to structure and represent content on webpages.

Is HTML a programming language?

No. HTML is a markup language.

What is HyperText?

HyperText is text that can contain links to other resources.

What is a tag?

A tag is markup syntax used to define or represent an HTML element.

Example:

```
<p>
```

What is an element?

An HTML element generally consists of a start tag, content, and an end tag.

Example:

```
<p>Hello</p>
```

Here the complete structure is an element.

What is an attribute?

An attribute provides additional information or configuration for an HTML element.

Example:

```
<input type="text">
```

Here **type** is an attribute.

Difference between **** and ****?

**** represents stylistically offset text, while **** represents strong importance.

Difference between `<i>` and `<em>`?

`<i>` represents alternate voice or mood, while `<em>` represents emphasis.

Difference between `<ol>` and `<ul>`?

`<ol>` represents an ordered list where sequence matters.

`<ul>` represents an unordered list where sequence generally does not matter.

Difference between `<th>` and `<td>`?

`<th>` represents a table header cell.

`<td>` represents a table data cell.

Difference between `rowspan` and `colspan`?

`rowspan` makes a cell span multiple rows.

`colspan` makes a cell span multiple columns.

What is the purpose of `alt`?

It provides alternative text for an image and improves accessibility and fallback information.

What is the purpose of `href`?

It specifies the destination of a hyperlink.

What is the purpose of `src`?

It specifies the source resource for elements such as images, audio, and video.

What is the purpose of `placeholder`?

It provides temporary hint text inside a form control.

Difference between radio button and checkbox?

Radio buttons are generally used for choosing one option from a group, while checkboxes allow independent selections and can allow multiple choices.

What is the purpose of `required`?

It indicates that a form control must have an appropriate value before normal form submission can proceed.

What is the purpose of `thead`, `tbody`, and `tfoot`?

They semantically group the header, main body, and footer portions of a table.
